

PS-3 Solⁿ outline.

Q)

1) Bernoulli Utility fⁿ: $v = \ln C$.

initial wealth: C_0 .

Invest C' , remaining wealth = $C_0 - C'$.

C' can become $2C'$ or $\frac{C'}{2}$.

At the end, we have either $(C_0 - C' + 2C') = C_0 + C'$
or $(C_0 - C' + \frac{C'}{2}) = C_0 - \frac{C'}{2}$.

Expc. Utility = $\frac{1}{2} \ln(C_0 + C') + \frac{1}{2} \ln(C_0 - \frac{C'}{2})$

Max^m with respect to C' ,

$$\frac{1}{2} \frac{1}{C_0 + C'} = \frac{1}{4} \frac{1}{C_0 - \frac{C'}{2}}$$

$$\Rightarrow 4(C_0 - \frac{C'}{2}) = 2(C_0 + C')$$

$$\Rightarrow 4C_0 - 2C' = 2C_0 + 2C'$$

or. $\odot 4C' = 2C_0$

$$\Rightarrow C' = \frac{1}{2} C_0$$

Note: Elasticity of X with respect to Y .

$$= \frac{\% \text{ change in } X}{\% \text{ " " } Y} = \frac{dX/X}{dY/Y} = \frac{dX}{dY} \cdot \frac{Y}{X}$$

Using this defⁿ,

$$\text{elasticity of } c' \text{ w.r.t. } c_0 = \frac{dc'}{dc_0} \times \frac{c_0}{c'}$$

$$= \frac{1}{2} \times 2 = 1.$$

Note: try with $u(c) = -e^{-\alpha c}$ (CARA).

2)

2) Markowitz's Bullet with -ve correlation.

The hyperbola is associated with the following eqn:

$$\mu = a\mu_1 + (1-a)\mu_2$$

$$\sigma^2 = a^2\sigma_1^2 + (1-a)^2\sigma_2^2 + 2a(1-a)\rho\sigma_1\sigma_2$$

If $\rho = -1$, then

$$\frac{d\sigma^2}{da} \Big|_{a=1, \rho=-1} = 2(\sigma_1^2 - \rho\sigma_1\sigma_2) \Big|_{\rho=-1} = 2(\sigma_1^2 + \sigma_1\sigma_2) > 0$$

$$\sim \text{ly, } \frac{d\sigma^2}{da} \Big|_{a=0, \rho=-1} = 2(-\sigma_2^2 + \rho\sigma_1\sigma_2) \Big|_{\rho=-1} = -2(\sigma_2^2 + \sigma_1\sigma_2) < 0.$$

So, at the two extremes, the slopes are +ve and -ve, respectively.

$$\text{Again, } \sigma^2 = (a\sigma_1 - (1-a)\sigma_2)^2.$$

$$\text{From here, if we put } a = \frac{\sigma_1}{\sigma_1 + \sigma_2} \Rightarrow \sigma^2 = 0$$

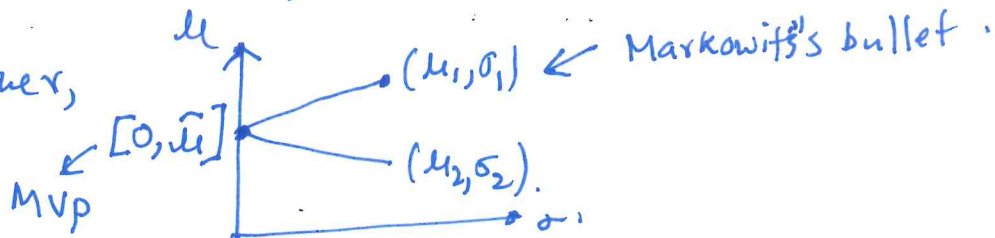
∴ it is possible to have a 0 risk portfolio.

$$\text{In general; } \frac{d\sigma^2}{da} = 2 \cdot \sigma [\sigma_1 + \sigma_2]$$

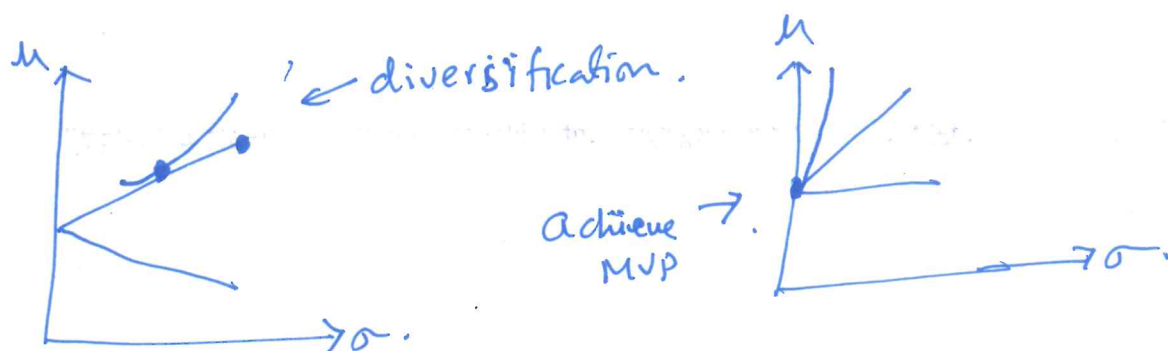
⇒ both μ and σ^2 varies monotonically with a .

⇒ The 'bullet' is a straight line.

All facts put together,



⇒ You can have diversification with $\rho = -1$, ~~but~~ or.
achieve MVP.



3). (i) Find μ and σ of L1 and L2.

$$\mu = E(x) = 0.2 \times 2 + 0.8 \times 12.$$

$$\sigma^2 = E(x - \mu)^2 = 0.2 (2 - \mu)^2 + 0.8 (12 - \mu)^2.$$

Similarly for the other lottery.

Then Compare EU of the above lotteries with the function $u = \ln W$.

[Hint: you will see one lottery is dominating the other in mean-variance sense.

However, the EU ranking is opposite]

4) $W_0 = 100$

$$W_0 - d = 70 \Rightarrow d = 30.$$

$$p = \frac{2}{3}, \quad 1 - p = \frac{1}{3} \text{ (prob. of accident)}$$

$$\text{With Insurance} \quad \alpha = \frac{I}{3} \text{ (fair insurance)}$$

$$\text{With insurance, } W_1 = 100 - \alpha = 100 - \frac{I}{3}$$

$$\begin{aligned} W_2 &= 70 - \alpha + I = 70 - \frac{I}{3} + \frac{I}{3} \\ &= 70 + \frac{2}{3} I \end{aligned}$$

The agent maximises $\frac{2}{3} \ln(100 - \frac{I}{3}) + \frac{1}{3} \ln(70 + \frac{I}{3})$

- In the second case, the agent maximises

$$\frac{2}{3} \sqrt{100 - \frac{I}{3}} + \frac{1}{3} \sqrt{70 + \frac{2}{3} I}$$

- Do you see any difference in α^* ?

5. Same as above. Now

$$\alpha = 1.25 \times \frac{1}{3} \times I$$

$\Rightarrow$ ~~see~~ repeat Qⁿ 4. Do you see any difference?

6. India wins: utility $\ln(w)$.

AOT wins

utility $\frac{1}{2} \ln(w)$: state dependent utility.
(utility terms)

[The $\frac{1}{2}$ is the 'weight' on your wealth if other team wins.
• If this is close to 1, you view Indian's win and AOT's win "at par"]

$$W_0 = 1000.$$

$$\text{AOT wins} = 1000 - x.$$

$$\text{India wins} = 1000 - x + 2x = 1000 + x.$$

Then the exercise is

$$\max_x \frac{1}{2} \ln(1000+x) + \frac{1}{2} \ln(1000-x)$$

I leave it as a trivial exercise.

[Interesting: do not use specific values. Let the

probabilities be $p, 1-p$, the bookie offering you

$m\alpha$ ($m > 1$) if your team wins and let the state-
dep. utility be $\alpha \ln x$ ($\alpha < 1$) if your team loses.

Then you maximise

$$\max_x p \ln(1000 + (m-1)x) + (1-p) \cdot \alpha \ln(1000-x)$$

See how x^* depends on various parameters.

7. Let the ~~probabi~~ utility of death be "0".

If not treated, his utility is

$$p_0 \cdot v(c) + (1-p_0) \cdot 0$$

$$= p_0 v(c)$$

If he spends on the physician, his utility is

$$= p v(c-z) + (1-p) \cdot 0 = p_0 v(c-z)$$

Thus he will spend as long as.

$$p v(c-z) \geq p_0 v(c) \quad | \quad p > p_0.$$

$\Rightarrow$ This defines an indifference relation between p and z .

(trade off: higher $p \Rightarrow$ ^{higher} ~~lower~~ $z \Rightarrow$ low $(c-z)$)

$u(p, z) = p v(c-z) = \bar{v} = p_0 v(c) \Rightarrow$ the max^m z I am willing to spend.

$$MRS = \frac{dz}{dp} = \frac{v(c-z)}{p v'(c-z)} > 0 \quad \text{if } v' > 0.$$

$\Rightarrow$ Extra amt that I am willing to pay given a small Δ in probability.

• Then find ~~the~~ how the slope changes with p ? That is

$$\frac{d}{dp} \left[\frac{dz}{dp} \right] > 0, < 0 \text{ or } = 0?$$

Implicitly differentiate MRS and find out.